

STAT 339

TUTORIAL SET 2

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

March, 2022

1. Examine the consistency of the following system of equation and find out if it is consistent.

Thus solve the equations:

I.

$$-1/x + 3/y + 4/z = 30$$

$$3/x + 2/y - 1/z = 9$$

$$2/x - 1/y + 2/z = 10$$

II.

$$x + y + 2z = 9$$

$$2x + 4y - 3z = 1$$

$$3x + 6y - 5z = 0$$

III.

$$2x - y - z = 2$$

$$x + 2y + z = 2$$

$$4x - 7y - 5z = 2$$

2. For what value of κ , will the equation have infinite solutions.
Solve for each case.

$$x + y + z = 1$$

$$2x + y + 4z = \kappa$$

$$4x - 7y - 5z = \kappa^2$$

3. Determine for what value of λ and μ , will the following equations have:

I. No Solution

II. Unique Solution

III. Infinite Many Solutions

The equations are as follows:

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 0 \\x + 2y + \lambda z &= \mu\end{aligned}$$

4. Determine the values of the constant λ and μ for which the system

$$\begin{aligned}\lambda x + 4y + 2z &= \mu \\4x + 7y + 3z &= 8 \\2x + 2y + z &= 2\end{aligned}$$

has: I. No Solution
II. Unique Solution
III. Infinite Many Solutions

5. Show that every square matrix A can be written as the sum of a symmetric matrix and skew-symmetric matrix. Illustrate this by writing the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 5 & -1 \\ 7 & -6 & 5 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

Hint: Consider $\frac{1}{2}[A + A^T]$ and $\frac{1}{2}[A - A^T]$

6. Solve the system of equations:

$$\begin{aligned}2x_1 - 3x_2 - x_3 + x_4 &= 0 \\3x_1 + 4x_2 - 4x_3 - 3x_4 &= 0 \\17x_2 - 15x_3 - 9x_4 &= 0\end{aligned}$$

Show that there is one solution of the system above that also satisfies the system of equation

$$\begin{aligned}x_1 + x_2 + x_3 + x_4 + 1 &= 0 \\x_1 - x_2 - x_3 - x_4 - 3 &= 0\end{aligned}$$

7. Show that the system of linear equations

$$\begin{aligned}2x + y + z &= -6\beta \\ \gamma x + 3y + 2z &= 2\beta \\ 2x + y + (\gamma + 1)z &= 4\end{aligned}$$

has a unique solution except when $\gamma = 0$ and when $\gamma = 6$. If $\gamma = 0$, prove that there is only one value of β for which a solution exists and find the general solution in this case.

Discuss the situation when $\gamma = 6$.

What are the conditions for inconsistency ?

8. Find all solutions of

$$2x_1 - 3x_2 - 7x_3 + 5x_4 + 2x_5 = -2$$

$$x_1 - 2x_2 - 4x_3 + 3x_4 + x_5 = -2$$

$$2x_1 - 4x_3 + 2x_4 + x_5 = 3$$

$$x_1 - 5x_2 - 7x_3 + 6x_4 + 2x_5 = -7$$